Frank (Jiawei) Zhang

+1(617) 599-0738 • jiaweizhang@gse.harvard.edu • Cambridge, MA, US • LinkedIn

EDUCATION

Harvard University — Cambridge, MA
Ed.M. in Learning Design Innovation and Technology — Aug 2025 – Present

Massachusetts Institute of Technology (MIT) — Cambridge, MA
Cross-Registered at *MIT Media lab*-AI Venture Studio *& MIT Sloan School of Management* — Sept 2025 – Present

UBC Sauder School of Business — Vancouver, BC
Master of Business Analytics, CGPA: 3.70/4.00 (>80%) — Aug 2023 - Aug 2024

University of Toronto — Toronto, ON
Bachelor of Digital Enterprise Management, CGPA: 3.74/4.00 (High Distinction) — Aug 2019 - Jun 2023

RESEARCH EXPERIENCE

Harvard Graduate School of Education) — Feb 2026 – Present
***Research Assistant*, Learning Innovation and Technology Lab** — Cambridge, MA

- Conducted an independent research project on individual differences in makerspace learning using data from 3 cohorts (56 students), integrating multimodal behavioral, interaction, and survey data into a unified student-level dataset for longitudinal analysis.
- Designed a theory-driven analytical framework based on Bandura's self-efficacy model, applied K-means clustering, and performed ANOVA/t-tests to identify distinct student profiles and explain how behavioral patterns related to changes in self-efficacy and sense of belonging.
- Validated data quality and analytical robustness by tracing raw variables back to source files, identifying a calculation inconsistency in one cohort's gain metrics, and reassessing results to ensure reliable evidence-based interpretation.

Tsinghua Shenzhen International Graduate School (Tsinghua-SIGS) — Sept 2024 – Aug 2025
Research Assistant — Shenzhen, China

- Developed optimization algorithms for the "OldPilot" system, improving elderly mobility and safety through personalized route planning and real-time health monitoring.
- Contributed to the design and implementation of spatiotemporal multimodal decision agents, optimizing path safety and efficiency under dynamic constraints.
- Led field testing and iterative improvements to refine system functionalities.

UsingAI — Aug 2024 – Mar 2025
AI Engineer — Shenzhen, China

- Enhanced YOLOv8 for real-time splitter port detection, achieving 98.55% accuracy in high-concurrency environments.
- Assisted in designing lightweight network architecture and attention mechanisms to enhance detection precision and reduce processing times to millisecond levels.
- Ensured successful implementation in data preprocessing, model training, and fine-tuning through team collaboration.

PUBLICATIONS/PRESENTATIONS

[1] J. W. Zhang, G. Wang and S. Zhi. "Improved YOLOv8 real-time object detection algorithm." Submitted to MDPI.Electronics (ISSN 2079-9292), 2024.
[2] J. W. Zhang, Y. Liang, T. Liang, and Y. Wang. "Predicting Optimal Pricing for Airbnb Listings at New York City Using Machine Learning.", 2024.
[3] J. W. Zhang. "Enhancing Content Engagement and Monetization on Social Media with ML Models", 2024.
[4] J. W. Zhang. "Optimizing Supply Chain Hubs for Global Humanitarian Logistics Using Network Flow and Linear Programming.", 2024.
[5] J. W. Zhang, J. Yang, L. Li, W. Zhou, and W. Wu. "Optimizing Airline Revenue with Simulation Modeling.", 2024.
[6] J. W. Zhang, X. Zhang, X. Lyu, Y. Yang, Y. Wang, and Z. Gu. "Optimizing U.S. Investment Portfolios Using CAPM and Linear Programming: A Debt-to-Equity Analysis Across Industries", 2024.
[7] J. W. Zhang, X. Shen, Y. Wang, and Y. Wang. "Optimizing Coffee Bean Sales Strategy for B2C Company-Bean Symphony-Through Data Analytics and Visualization.", 2023.
[8] J. W. Zhang, B. Yang, Y. Liu, Z. Gu, and X. Shen. "Optimizing Portfolio Management with Active Strategies: A Two-Year Financial Modeling Case Study.", 2023.
[9] J. W. Zhang, S. Ahmed, Y. Ding, H. Xia, and B. Zhao. "Inventory Management at JOANN Using JIAT: A Data-Driven Strategy for Seasonal Demand.", 2023.
[10] J. W. Zhang, S. Ahmed, Y. Ding, H. Xia, and B. Zhao. "Tackling the Planning Fallacy in NYC Parks & Recreation: Data-Driven Solutions for Budget and Schedule Overruns.", 2023.

WORKING EXPERIENCE

Wyloo Metals — Toronto, CA

Data Analyst Co-op May 2024 - Aug 2024

- Utilized Agile project management to oversee document architecture for chatbot and enterprise search initiatives, delivering a Minimum Viable Product (MVP) plan with optimized resource allocation and risk mitigation strategies.
- Developed a Python-based automation system for meeting documentation, enhancing operational efficiency by 50% and providing critical data for AI tool development.
- Collaborated with cross-functional teams to resolve data quality issues in Databricks, aligning data governance criteria.

HireBeat New York, US

Business & Data Analyst Intern Jul 2023 - Nov 2023

- Developed Power BI dashboards to track employee engagement and turnover, identifying trends and enabling targeted retention strategies that reduced turnover by 15%.
- Conducted gap analysis to define user needs and deliver a performance review feature through Agile sprints in JIRA, collaborating with HR and product teams. Launched a self-assessment tool that cut review cycle time by 30%.
- Optimized the employee onboarding process by implementing a digital checklist, reducing onboarding time by 25%.

Accenture Strategy & Consulting Shanghai, China

Business Analyst Intern May 2022 - Jul 2022

- Built a sales performance dashboard using SQL and Power BI, automating data aggregation and reducing manual processing time by 40%, accelerating BI adoption across departments.
- Supported the design of an enterprise data warehouse, defining fact and dimension tables with client partners, and implementing Type 1 & 2 slowly changing dimensions to ensure robust business analysis.

EXTRACURRICULAR

Harvard President's Innovation Challenge-Innovation Lab Cambridge, MA

Team Lead Dec 2025 - Present

- Led and organized a cross-disciplinary team from financial experts to engineers to define problem scope, user needs, and venture positioning for an AI-driven innovation project. Drove end-to-end concept development, from problem framing and prototyping to pitch refinement for competition review.

MIT Media lab AI Venture Demo Day, Shortlisted Cambridge, MA

Team Lead Aug 2024 - Dec 2025

- Led an AI venture from concept to demo, integrating agentic system design, user feedback, and rapid prototyping.
- Shortlisted among peer teams; refined technical narrative and product strategy for public Demo Day presentation.

Google Hackathon 2024, Third Place Winner Vancouver, Canada

Team Lead Jan 2023 - Mar 2024

- Designed an E-commerce dashboard integrating real-time analytics for web, customer, product, and logistics data using BigQuery, simplifying complex datasets into actionable insights.
- Consolidated multi-source data into a unified warehouse to enhance cross-platform analysis, improving efficiency.
- Leveraged AI for hyper-personalized marketing, automating real-time insights generation, and democratizing data usage for both technical and non-technical stakeholders via integrating Gmail, Google Analytics, and Gemini.

UBC Sauder Student Council (MBAN) Vancouver, Canada

Vice President of International Aug 2023 - May 2024

- Supported 100+ international students by holding international student seminars for smooth cultural transitions.
- Coordinated major cultural events and career sessions, fostering community engagement and professional growth.
- Mediated work overload and conflicts during high-pressure periods by planning extra-curriculum activities, such Beach outings, hiking trip, and boxing work out sessions, facilitating better mental health and student academic performance.

SKILLS AND ADDITIONAL INFORMATION

- **Programing:** Python, SQL, R, Processing(Java), Excel VBA, Web(HTML,CSS, Javascript), Arduino, AI/ML.
- **Tools:** JIRA, Power BI, Tableau, Mango DB, CQL, Google(SEO/Looker/BigQuery), Adobe(Ps/Ai/An/Pr/Coreldraw).
- **Certificate**: Project Management Essentials Certified (PMEC), Bloomberg BFF&BMC, Accenture Data Analytics.